

| Question              | Marking Guidance                                                                                                                                                                                                                                                                                                                       | Mark                                                                        | Comments                                                         |                                                                             |                                   |       |      |     |      |   |                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------|-----------------------------------------------------------------------------|-----------------------------------|-------|------|-----|------|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8.1                   | <table border="1"> <tr> <th>Water potential / MPa</th><th>Concentration of sodium chloride solution / mol dm<sup>-3</sup></th><th>Volume of 1 mol dm<sup>-3</sup> sodium chloride solution / cm<sup>3</sup></th><th>Volume of Water / cm<sup>3</sup></th></tr> <tr> <td>-1.95</td><td>0.04</td><td>0.8</td><td>19.2</td></tr> </table> | Water potential / MPa                                                       | Concentration of sodium chloride solution / mol dm <sup>-3</sup> | Volume of 1 mol dm <sup>-3</sup> sodium chloride solution / cm <sup>3</sup> | Volume of Water / cm <sup>3</sup> | -1.95 | 0.04 | 0.8 | 19.2 | 2 | 1 mark for each row.<br><br>If values do not match the given unit, max 1.<br><br>Accept dm <sup>3</sup> / mm <sup>3</sup> for volume unit.<br><br>Accept 0.0008/8 x 10 <sup>-4</sup> and 0.0192/1.92 x 10 <sup>-2</sup><br><br>Accept 800 and 19200<br><br>Ignore units in 2 <sup>nd</sup> row.<br><br>Do not accept mm <sup>-3</sup> /cm <sup>-3</sup> /dm <sup>-3</sup> /ml |
| Water potential / MPa | Concentration of sodium chloride solution / mol dm <sup>-3</sup>                                                                                                                                                                                                                                                                       | Volume of 1 mol dm <sup>-3</sup> sodium chloride solution / cm <sup>3</sup> | Volume of Water / cm <sup>3</sup>                                |                                                                             |                                   |       |      |     |      |   |                                                                                                                                                                                                                                                                                                                                                                               |
| -1.95                 | 0.04                                                                                                                                                                                                                                                                                                                                   | 0.8                                                                         | 19.2                                                             |                                                                             |                                   |       |      |     |      |   |                                                                                                                                                                                                                                                                                                                                                                               |
| 8.2                   | Correct answer of 0.07 (mol dm <sup>-3</sup> ) = <b>2 marks</b> ;;<br>Incorrect answer 1 mark for any evidence of 48.6 to 48.8<br><b>OR</b><br>0.02<br><b>OR</b><br>0.7<br><b>OR</b><br>A final answer between 0.04 and 0.10<br><b>OR</b><br>A final answer of <b>minus</b> 0.07/-0.07;                                                | 2                                                                           | Ignore minus signs on other 1 mark options.                      |                                                                             |                                   |       |      |     |      |   |                                                                                                                                                                                                                                                                                                                                                                               |

| Question | Marking Guidance                                                                                                                                                                                                                                                                                               | Mark  | Comments                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8.3      | <p>Correct answer of 9 (cm<sup>2</sup>) = <b>2 marks</b>;;</p> <p>Incorrect answer 1 mark for evidence of water potential of between -1.85 and -1.95 (MPa)</p> <p><b>OR</b></p> <p>growth of 15%</p> <p><b>OR</b></p> <p>69 (cm<sup>2</sup>)</p> <p><b>OR</b></p> <p>A final answer between 8.7 and &lt;9;</p> | 2     | <p>Allow 9.0</p> <p>Accept correct reading labelled on the graph shown on <b>Figure 8</b> or <b>Figure 9</b>.</p>                                                                                                                                                                                                                                                                                                                   |
| 8.4      | <p><b>EITHER</b></p> <p>1. Low/slow growth;</p> <p>2. Due to smaller number/area of stomata (for gas exchange);</p> <p><b>OR</b></p> <p>3. Growth may continue at lower water potentials;</p> <p>4. (Due to) adaptations in enzymes involved in photosynthesis/metabolic reactions;</p>                        | 2 max | <p>Mark as pair – 1 and 2 <b>OR</b> 3 and 4.</p> <p>2. Reference to stomata must not relate only to water loss.</p>                                                                                                                                                                                                                                                                                                                 |
| 8.5      | <p>1. Stomata close;</p> <p>2. Less carbon dioxide (uptake) for less photosynthesis/glucose production;</p>                                                                                                                                                                                                    | 2     | <p>2. 'Less' only required once.</p> <p>2. Reject 'no photosynthesis' but accept 'carbon dioxide can't enter so less photosynthesis'.</p> <p>2. Ignore oxygen for respiration but reject oxygen for photosynthesis.</p> <p>2. Ignore less water for photosynthesis.</p> <p>2. Accept only correct chemical formulae.</p> <p>2. For 'glucose' accept named product of photosynthesis eg triose phosphate, TP, amino acid, lipid.</p> |

